LONGEVITY 15

Rapamycin (sirolimus)

mTOR inhibitor; a prescription immune-suppressing medicine



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
mTOR inhibitor	Transplant, LAM, aging biology	Large trials; FDA approved	FDA approved (Rapamune) · aging use off-label

STUDIED FOR

Kidney transplant — HUMAN	LAM lung disease — HUMAN	Body composition — HUMAN	
Fitness, older adults — HUMAN	Lifespan — ANIMAL		

Lifespan data are from mice. Aging trials in people were small, 8 to 48 weeks long. None measured how long people live.

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved since 1999 (Rapamune)
- Aging: small randomized trials, 25 to 129 people
- No Phase 3 aging trial found

NOT YET ESTABLISHED

- Effect on aging in people
- Long-term use in healthy people

HOW IT WORKS IN THE BODY

Sirolimus ▶ Binds FKBP-12 ▶ Inhibits mTOR ▶ T-cell growth

- **Binds FKBP-12;** the complex binds and inhibits mTOR, a key regulatory kinase (label)
- **Suppresses T-cell proliferation** and antibody production (label)
- **Extended lifespan** in mice fed it from about 600 days of age
- **In people,** an effect on aging is not established

QUICK PRACTICAL INFO

HALF-LIFE About 62 hours (transplant patients, label)	ROUTES STUDIED By mouth (tablets, solution); also IV (Fyarro) and gel (Hyftor)
PHARMACOKINETICS Peak about 1 to 2 hours (solution); CYP3A4 and P-gp substrate	SOLD AS Prescription (Rapamune, generics); compounded in one aging trial
MONITORED IN TRIALS Sirolimus trough levels, blood fats, kidney protein, infections	SPORT (WADA) Not on the 2026 list

REGULATORY

FDA	Approved 1999, transplant and LAM; aging off-label
FDA	Boxed warning: infection, lymphoma, other cancers
EMA	Rapamune authorised EU-wide since 2001
WADA	Not listed

REFERENCES

1. FDA, sirolimus tablets prescribing information (DailyMed), Mar 2026
2. Nature, rapamycin fed late in life extends lifespan in mice, Jul 2009
3. WADA, 2026 Prohibited List

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